

Kasra Siavashpour

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Education

Sharif University of Technology (SUT), Tehran, Iran

Bachelor of Science, Computer Science

2020 – Jun 2026

Allameh Tabatabaee High School (selective, non-governmental), Tehran, Iran

High School Diploma, Mathematics and Physics

2016 – 2020

Research Interests

- Distributed & Decentralized Systems
- IoT & Cyber-Physical Systems
- Multi-Agent Systems
- AI-Driven Network Automation

Publications

Under Review

- A. Vaezi, S. Niknejad, **K. Siavashpour**, A. Dadras, “FAIR: A Novel Envy-Free Multi-Agent Model for Traffic Systems,” under review at the International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2027

In Preparation

- A. Siavashpour, **K. Siavashpour**, M. Azadegan, B. Safaei, H. Peyvandi, M. Imran, “A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems,” manuscript in preparation for submission to the IEEE Open Journal of the Communications Society (OJ-COMS)
- A. Vaezi, A. Daneshgar, A. Dadras, **K. Siavashpour**, “The Loop of the Rings: A Fully Decentralized Reliable Cooperative System,” manuscript in preparation for submission to the IEEE Symposium on Foundations of Computer Science (FOCS) 2027, [[Preliminary version on arXiv](#)]

Research Experience

Research Assistant, Institute for Research in Fundamental Sciences (IPM)

& **Sharif University of Technology**, Tehran, Iran

supervisor: [Dr. Arash Vaezi](#)

May 2026 – Present

project: “FAIR: A Novel Envy-Free Multi-Agent Model for Traffic Systems”

- Co-developed the paper’s revised proofs after simulation on real networks contradicted them.
- Designed and built the evaluation: a self-contained simulator run on seven city road networks.
- Found congestion stays bounded under sustained demand, where shortest-path routing diverges.
- Showed the fairness gain falls on the worst-served trips, leaving the median almost unchanged.

Research Assistant, Institute for Research in Fundamental Sciences (IPM)

& **Sharif University of Technology**, Tehran, Iran

supervisors: [Dr. Arash Vaezi](#), [Dr. Amir Daneshgar](#)

Aug 2024 – Present

project: “The Loop of the Rings: A Fully Decentralized Reliable Cooperative System”

- Implemented the LoR core in Go: rounds, checkpoints, verification teams and TCB updates.
- Modeled the randomized protocol and swept α , β , p against verification accuracy and cost.
- Validated the theory with large-scale adversarial simulations, reproducing its key thresholds.
- Built a reproducible Go CLI with structured logs and plots for experiment reporting.

Research Assistant, Sharif University of Technology, Tehran, Iran

Aug 2024 – Present

supervisors: [Dr. Hossein Peyvandi](#), [Dr. Masoumeh Azadegan](#) (Khatam University)

collaborators: [Dr. Bardia Safaei](#), [Dr. Muhammad Imran](#) (University of Glasgow)

project: “A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems”

- Surveyed 114 papers on Self-Organizing Networks (SON) for 5G and 6G cellular systems.
- Mapped each SON function to its control parameters, performance objectives and algorithms.
- Compared optimization, heuristic and learning-based methods on scalability and robustness.
- Identified open problems, including conflicts between SON functions and SON’s role in 6G.

Research Assistant, Politecnico di Milano, Milan, Italy (Remote)

supervisor: [Dr. Gianni Antichi](#)

May 2024 – Aug 2024

- Researched WebAssembly for systems acceleration and designed a benchmarking plan for CPU, memory, latency, and throughput.

Research Assistant, Sharif University of Technology, Tehran, Iran

supervisor: [Dr. Bardia Safaei](#)

Oct 2023 – Jan 2024

- Researched nature-inspired FANET and IoT cluster routing with mobility-aware benchmarking.
- Prototyped beaconing and neighbor discovery, and analyzed energy-delivery trade-offs.
- Built reproducible pipelines (configs, seeds, scripts) for experiment reruns.

Honors & Awards

Top 0.1% among 155,250 candidates, Iranian University Entrance Exam (B.Sc.)	<i>2020</i>
1 st place in National Hovercraft Competition (two consecutive years), design & control	<i>2016 – 2017</i>
1 st place in Provincial Chess Competition (High School)	<i>2016</i>
1 st place in Provincial Table Tennis Competition (High School)	<i>2015</i>

Selected Courses

- | | |
|--|-------------------------------|
| o Machine Learning Systems Design: 20/20 | o Image Processing: 19/20 |
| o Data and Network Security: 19.6/20 | o Numerical Analysis: 18.5/20 |
| o Stochastic Processes: 19.2/20 | |
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Teaching Experience

Teaching Assistant, Sharif University of Technology

Computer Networks, Dr. Mahdi Dolati	<i>Fall 2026, Winter 2026</i>
Operating Systems, Dr. Hossein Asadi	<i>Fall 2026</i>
Operating Systems, Dr. Hossein Asadi & Dr. Rasul Jalili	<i>Fall 2024</i>
Computer Networks, Dr. Hossein Peyvandi	<i>Winter 2024</i>
Operating Systems, Dr. Arash Vaezi [Piazza]	<i>Winter 2024</i>
Computer Networks, Dr. Kambiz Mizanian	<i>Fall 2023</i>
Operating Systems, Dr. Mohaddeseh Mirbeygi & Dr. MohammadAli Mirzaei	<i>Fall 2023</i>

Designed and graded weekly assignments (120+ students in the largest classes), wrote rubrics, held office hours, guided projects and built autograders.

Notable Projects

SliceDB: Distributed Key-Value Store (*Go, etcd*) [\[GitHub Link\]](#) *2025 – 2026*

- Built a sharded key-value store with write-ahead-log replication and leader election through etcd.
- Added automatic failover and online resharding; killing a node under load caused 0 failed requests.

Multimodal Product Search (*Python, CLIP, FAISS*) [\[GitHub Link\]](#) *2025 – 2026*

- Built text and image search over 1M products, evaluated on 5,000 real user queries.

Highly Available HDFS + Spark (*Hadoop, ZooKeeper, Spark; team of two*) [\[GitHub Link\]](#) *2025*

- Deployed HDFS with automatic NameNode failover and ran Spark queries on NYC taxi data over it.

Go Judge: Online Judge (*Go, PostgreSQL, Docker; team of two*) [\[GitHub Link\]](#) *2025*

- Built the web server, UI and database schema of a judge that runs each submission in a Docker sandbox.

Go Download Manager (*Go; team of two*) [\[GitHub Link\]](#) *2025*

- Built a terminal downloader with parallel range requests, pause/resume and per-queue speed limits.

Skills	Programming Languages: Java, Python, C/C++, Go, JavaScript, Bash Tools: Git, Docker, L ^A T _E X, NS-3, PostgreSQL, Spark Operating Systems: Ubuntu, Windows, macOS	Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, Matplotlib Web Programming: Vue, Flask, HTML5, CSS
Languages	Persian (Native) English (C1)	